

ELECTRIC DOOR LOCK SYSTEM

Project Report

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Batch - G

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Abstract

The Electric Door Lock System presented in this project demonstrates a reliable, low-cost, and secure way of controlling access using an OTP-based verification process through GSM communication. Unlike Wi-Fi-based locks, this system operates independently of internet connectivity, ensuring consistent performance in diverse environments. The combination of Arduino microcontroller, GSM module, servo motor, keypad, and LCD provides a compact and scalable design.

This report outlines the theoretical background, hardware implementation, software logic, and testing of the system. It also discusses potential enhancements, future applications, and the role of IoT in next-generation security solutions.

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1. Introduction

Security remains one of the most fundamental and enduring concerns of human civilization, forming the bedrock upon which personal safety, property, and societal trust are constructed. The assurance of security within one’s residence or office is not merely a luxury, but a core necessity. For centuries, the primary instrument of this security has been the mechanical lock and key. This system, while simplistic, reliable in its own right, and ubiquitous, presents as an anachronism in an increasingly connected digital world. Its security is purely physical, and its vulnerabilities are stark and well-documented: a physical key can be easily lost, stolen, or duplicated without authorization by any individual with brief access to it.

This outdated key management paradigm creates an immense logistical burden. The loss of a key is not a mere inconvenience; it constitutes a permanent, unknown vulnerability that necessitates the costly and time-consuming process of replacing the entire lock cylinder. For a homeowner, this represents a source of significant anxiety and expense; for a landlord or business owner, re-keying locks for new tenants or following an employee’s departure is a significant, recurring operational cost. This disparity—this gap between static, physical security paradigms and the dynamic nature of digital life—has created a clear and urgent demand for a more intelligent, flexible, and truly secure solution.

The rise of the Internet of Things (IoT) has sought to address this demand, ushering in the era of the “smart lock.” This evolution, however, has introduced a new and complex set of trade-offs, often exchanging established, physical vulnerabilities for novel, digital counterparts. The first wave of common smart locks, such as those utilizing Bluetooth Low Energy (BLE) or Radio-Frequency Identification (RFID), offered a minor convenience. They successfully replaced the physical key with a smartphone application or a key fob. While this solved the problem of mechanical key duplication, it failed to resolve the core challenges of operational range and remote access. These locks still operate on a “digital leash,” requiring the user to be in close physical proximity to the door. Furthermore, the physical token (the RFID card or fob) is just as susceptible to loss or theft as a traditional key and, in many instances, can be “cloned” with relative ease, effectively replicating the original vulnerability within a new technological modality.

The second, more advanced wave of smart locks embraced Wi-Fi and cloud connectivity, finally delivering on the promise of true remote access. These systems are highly convenient, permitting a user to lock, unlock, and monitor their door from any location worldwide with an internet connection. They can grant temporary digital credentials to a guest remotely or verify the door’s locked status from afar. However, this convenience is accompanied by a significant caveat: a critical, multi-layered dependency on complex and frequently unreliable infrastructure. These locks are completely reliant on a functional chain of failure-prone components, each of which represents a potential point of failure:

the lock's Wi-Fi connection, the home router's stability, the internet service provider's uptime, and the continuous operation of a third-party cloud server. A simple power outage, a router malfunction, a local internet service interruption, or a server-side issue with the manufacturer can instantly render the lock inoperable, consequently forcing the user to revert to a mechanical backup key—the very device they sought to replace.

Furthermore, by connecting the core mechanism of home security directly to the public internet, these systems introduce a significant and dangerous digital "attack surface." This term denotes the multitude of entry points a malicious actor can exploit. A home's Wi-Fi network, if poorly secured, becomes a direct gateway to the lock. A compromised cloud account password, a vulnerability in the lock's own firmware, or a "man-in-the-middle" attack on the local network could potentially allow a remote attacker to bypass the lock entirely. This reliance on the internet creates a single, catastrophic point of failure, trading robust, simplistic security for a convenience that is fragile, complex, and fraught with digital risk.

This project directly confronts this critical gap: the need for a system that provides the remote accessibility of a Wi-Fi lock without its associated critical dependencies. This report proposes a novel solution that bridges this divide. Herein, we present the design, implementation, and evaluation of an Electric Door Lock System based on the Global System for Mobile Communications (GSM) network and One-Time Password (OTP) authentication. This architecture is designed to achieve the convenience of a remote system whilst mitigating its inherent dependencies.

This system is designed from its inception to be independent and resilient. Rather than relying on a user's local, failure-prone Wi-Fi network, our solution leverages the ubiquitous and robust GSM cellular network. This is the same mature, secure, and widespread infrastructure that powers global mobile communication, which is constructed with carrier-grade reliability and power redundancy far exceeding that of consumer-grade networking equipment. This design choice immediately isolates the lock from local internet and power outages, ensuring it remains operational provided a cellular signal is available.

The security of the system is anchored by the One-Time Password (OTP). Unlike a static password, which is a "static" credential that can be shared, stolen, or observed ("shoulder-surfing"), the OTP is a dynamic, "transient" credential. It is generated by the system for a single use and sent directly to the authorized user's phone via an "out-of-band" channel (SMS). This password is valid for only one access attempt and expires immediately after use. This "ephemeral" nature renders credential theft and replay attacks completely ineffective, thereby adopting a security standard trusted by the financial and banking industries for their most high-stakes transactions.

The motivation for this project is to create an access control solution that is not only secure and reliable but also affordable and accessible. By building the system around the open-source Arduino microcontroller platform and widely available, low-cost components, it is demonstrated that high-grade, independent security need not be a high-cost, proprietary luxury. This open-source approach arguably democratizes security, providing a blueprint that is replicable, auditable, and repairable, in stark contrast to the "black box" nature of many commercial systems.

This report provides a comprehensive documentation of the project's entire lifecycle, serving as a comprehensive guide to its design and function. It begins with a detailed Literature Review to situate this work within the current landscape of smart lock technologies. It subsequently outlines the complete System Design and Methodology, detailing the high-level architecture, workflow, and core working prin-

ciples. The following Hardware and Software Design sections provide a granular, technical breakdown of the components used and the logic implemented in the code, offering a replicable technical blueprint. Finally, the report presents the Testing and Results to empirically validate the claims of reliability and security, followed by a conclusion on the project's successful validation and a discussion of its potential for future scope and real-world applications.

Motivation

The motivation behind this project stems from the need for an independent, affordable, and reliable access control solution that works even without Wi-Fi. With GSM coverage widely available, integrating OTP-based verification into a GSM module ensures accessibility almost anywhere.

Advantages of GSM-based Security

- Works without local Wi-Fi or internet.
- Offers secure and encrypted communication.
- Provides global accessibility.
- Supports real-time SMS alerts for every action.

2. Objectives

Primary Objective

To design and develop a GSM-based electric door lock system that provides secure and convenient access control using OTP verification.

Secondary Objectives

- To implement the system using Arduino and other open-source components.
- To achieve independence from internet connectivity.
- To ensure strong protection against password reuse and interception.
- To create an intuitive interface using an LCD and keypad.
- To provide feedback and alerts to the user via SMS.

Problem Statement

Most existing smart locks either rely on physical tokens (RFID, keys) or constant internet access (Wi-Fi). Both are prone to failure under real-world conditions. A GSM-based OTP system provides reliability and security by leveraging cellular networks and dynamic password generation.

3. Literature Review

A comprehensive review of existing literature and commercial smart lock solutions reveals a market segmented by communication technology and credentialing methods. Each category presents a distinct set of trade-offs between convenience, security, and infrastructure dependency. The primary categories are: 1) Short-range proximity-based systems, 2) Networked IoT and cloud-based systems, and 3) Static credential systems.

3.1. Bluetooth and RFID Systems

This category includes common consumer-grade locks.

Bluetooth Low Energy (BLE): BLE systems use smartphones as digital keys, but are strictly limited to a short operational range (< 10 meters), offering no remote operation. They are also susceptible to signal interruption and application-layer vulnerabilities.

Radio-Frequency Identification (RFID) / Near Field Communication (NFC): Common in commercial settings, RFID/NFC systems use static credentials (cards or fobs) that function as digital keys. This makes them fundamentally vulnerable to being lost, stolen, or cloned.

3.2. Wi-Fi and Cloud-Based Systems

To address range limitations, Wi-Fi locks connect to the internet, offering remote access and activity logs. However, this convenience introduces a critical dependency on a complex chain of failure points: the local Wi-Fi, the ISP, and the manufacturer's cloud servers. A failure anywhere in this chain can render the lock inoperable. Furthermore, connecting the lock to the public internet creates a significant "attack surface" vulnerable to digital threats and raises data privacy concerns.

3.3. Static Credential Systems

The most traditional form of keyless entry, the static keypad lock, requires a single, unchanging password. While this eliminates the physical key, the credential's static nature is its primary flaw. This password can be easily shared, visually observed ("shoulder-surfing"), or guessed over time. An additional, often-overlooked vulnerability is the physical wear on the keypad itself, where the most-used numbers become smudged or worn, providing a strong visual clue to an attacker. These systems offer no remote monitoring and are only marginally more secure than a basic mechanical lock.

3.4. GSM-Based OTP Systems

A more robust category leverages the Global System for Mobile Communications (GSM) network, paired with a One-Time Password (OTP).

Infrastructure Independence: The lock operates as an independent node on the resilient cellular network, requiring no local Wi-Fi or internet.

Dynamic Credentialing (OTP): The system uses an "ephemeral" password, generated for a single use. This inherently mitigates replay attacks, "shoulder-surfing," and credential cloning.

Out-of-Band (OOB) Authentication: Delivering the OTP via SMS (one channel) while entering it on a keypad (a separate channel) makes interception significantly more difficult.

Summary

The literature review reveals a clear trilemma in existing smart lock solutions, forcing a trade-off between convenience (Wi-Fi), proximity-based reliability (BLE/RFID), and security (static credentials). Wi-Fi locks are convenient but unreliable and insecure. BLE/RFID locks are reliable for proximity use but lack remote features and are vulnerable to credential cloning. Static keypads are simple but insecure.

This analysis identifies a clear research and development gap for a solution that is simultaneously:

Remotely Accessible: Capable of being monitored and controlled from afar.

Infrastructure-Independent: Not reliant on failure-prone local Wi-Fi or cloud servers.

Highly Secure: Uses dynamic, single-use credentials to prevent unauthorized access.

The proposed GSM-based OTP system is specifically designed to fill this gap, offering a novel synthesis of the reliability of the cellular network with the proven security of dynamic password authentication.

4. System Design and Methodology

4.1. Overview

The proposed system combines hardware and software elements to provide a complete access control mechanism. The Arduino acts as the brain of the system, managing communication between input, processing, and output components.

4.2. System Components

- **Arduino UNO:** Serves as the controller for logic execution.
- **GSM Module (SIM900A/SIM800L):** Handles communication by sending and receiving SMS.
- **16x2 LCD Display:** Displays OTP prompts and system messages.
- **4x4 Keypad:** Used for user input.
- **Servo Motor:** Acts as the locking actuator.

4.3. Working Principle

The system follows this process:

1. Power on and initialize modules.
2. Generate a random OTP.
3. Send OTP to user via GSM.
4. Wait for keypad input.
5. Verify OTP. If correct, unlock door for 5 seconds.
6. If incorrect, send alert message to owner.

5. Block Diagram and Workflow

5.1. System Flow

1. System initializes GSM, LCD, and keypad.
2. Generates and sends OTP.
3. Waits for input from the user.
4. If OTP is correct, unlocks and relocks after a delay.
5. Sends SMS updates for all actions.

5.2. GSM Communication Flow

The GSM module uses AT commands for communication. For example:

- AT+CMGF=1 – Set SMS text mode.
- AT+CMGS="number" – Send message command.
- CTRL+Z – End SMS and transmit.

This sequence ensures that the system can reliably communicate with the owner.

5.3. Error Handling

If the GSM network fails, the system retains the last OTP and allows retry after reconnection. Incorrect OTP attempts trigger immediate SMS alerts.

OTP Smart Lock System (Inputs/Outputs)

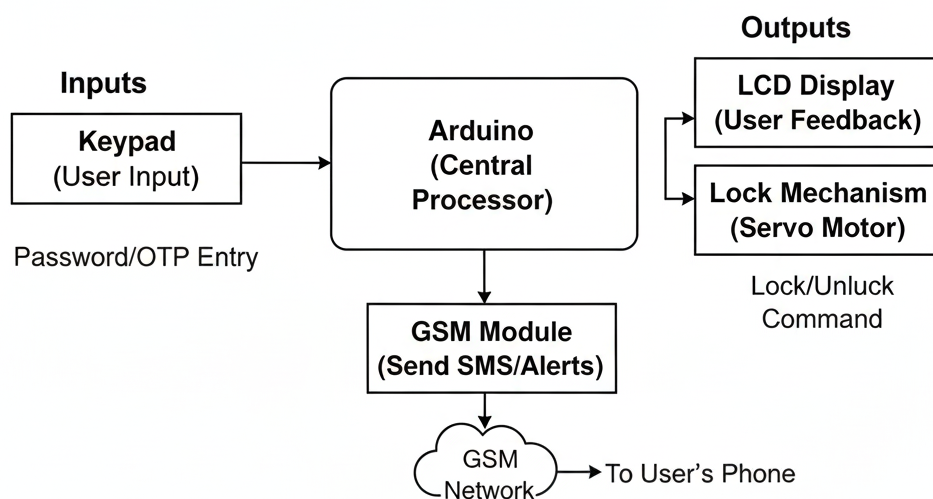


Figure 5.1: Block Diagram of the Electric Door Lock System

6. Hardware Design

6.1. Component Description

Arduino UNO

Based on the ATmega328P microcontroller, it provides digital I/O, timers, and serial communication. It manages OTP generation and verification.

GSM Module

Modules like SIM900A communicate over the 2G network using standard AT commands. It transmits messages and receives control inputs.

Keypad and LCD

The 4x4 keypad serves as a user interface, while the I2C-based LCD provides real-time feedback with minimal wiring.

Servo Motor

The SG90 servo operates at 5V and rotates between 0° (locked) and 90° (unlocked). It is ideal for prototype-level mechanical control.

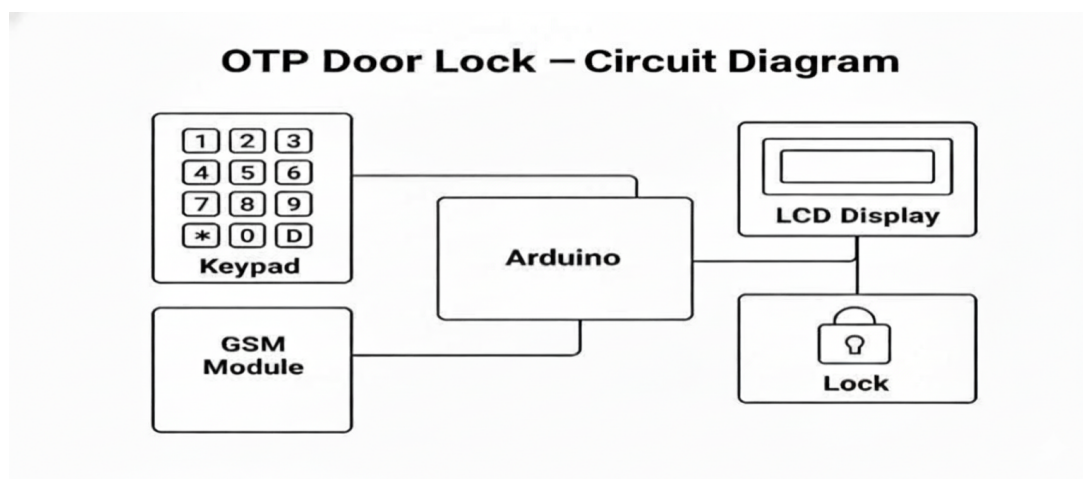


Figure 6.1: Circuit Diagram of the Electric Door Lock System

7. Software Design

7.1. Algorithm

1. Initialize system. 2. Check EEPROM for OTP flag. 3. Generate new OTP if required. 4. Send OTP using GSM. 5. Await input via keypad. 6. Compare input with OTP. 7. Unlock door if matched. 8. Send alert on failure.

7.2. Code

Listing 7.1: Arduino Code for GSM-based Electric Door Lock System

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Keypad.h>
#include <Servo.h>
#include <SoftwareSerial.h>
#include <EEPROM.h>

// LCD CONFIG
LiquidCrystal_I2C lcd(0x27, 16, 2);

// KEYPAD
const byte ROWS = 4;
const byte COLS = 4;
char rawKeys[ROWS][COLS] = {
  {'1', '2', '3', 'A'},
  {'4', '5', '6', 'B'},
  {'7', '8', '9', 'C'},
  {'*', '0', '#', 'D'}
};

byte rowPins[ROWS] = {2, 3, 4, 5};
byte colPins[COLS] = {6, 7, 8, 9};
Keypad keypad = Keypad(makeKeymap(rawKeys), rowPins, colPins, ROWS, COLS);

// SERVO CONFIG
Servo lockServo;
int servoPin = 10;

// GSM CONFIG
SoftwareSerial gsm(11, 12); // (RX, TX)
```



```
String phone = "+918334802800"; // PHONE NUMBER

// VARIABLES
String generatedPassword = "";
String inputPassword = "";
const int otpFlagAddress = 0; // EEPROM location to store flag

// KEYPAD FIX
char correctKey(char k) {
    switch (k) {
        case 'D': return '1';
        case 'C': return '2';
        case 'B': return '3';
        case 'A': return 'A';
        case '#': return '4';
        case '9': return '5';
        case '6': return '6';
        case '3': return 'B';
        case '0': return '7';
        case '8': return '8';
        case '5': return '9';
        case '2': return 'C';
        case '7': return '0';
        case '4': return '#';
        case '1': return 'D';
        default: return k;
    }
}

// GSM MESSAGE SENDING
void sendSMS(String msg) {
    gsm.println("AT+CMGF=1");
    delay(500);
    gsm.println("AT+CMGS=\"" + phone + "\"");
    delay(500);
    gsm.print(msg);
    delay(500);
    gsm.write(26); // to send
    delay(3000);
}

// OTP GENERATION
String generateOTP() {
    randomSeed(analogRead(0));
    int otp = random(1000, 9999);
    return String(otp);
}

void setup() {
    lcd.init();
    lcd.backlight();
    lcd.clear();
}
```

```
lcd.print("System Starting...");

lockServo.attach(servoPin);
lockServo.write(0);

gsm.begin(9600);
delay(3000);
gsm.println("AT");
delay(1000);

byte otpSentFlag = EEPROM.read(otpFlagAddress);

if (otpSentFlag != 1) {
    generatedPassword = generateOTP();
    sendSMS("Your Door Lock OTP is: " + generatedPassword);
    EEPROM.write(otpFlagAddress, 1);
    lcd.clear();
    lcd.print("OTP Sent to phone");
    delay(3000);
} else {
    lcd.clear();
    lcd.print("OTP already sent");
    delay(2000);
}

lcd.clear();
lcd.print("Enter Password:");
}

void loop() {
    char key = keypad.getKey();

    if (key) {
        char actual = correctKey(key);

        // * to clear
        if (actual == '*') {
            inputPassword = "";
            lcd.clear();
            lcd.print("Enter Password:");
        }

        // # to submit
        else if (actual == '#') {
            lcd.clear();
            if (inputPassword == generatedPassword && generatedPassword != "") {
                lcd.print("Access Granted");
                sendSMS("Door unlocked with correct OTP.");
                lockServo.write(90);
                delay(5000);
                lockServo.write(0);
            } else {
```

```
        lcd.print("Wrong Password");
        sendSMS("Alert: Wrong password attempt!");
    }
    delay(2000);
    lcd.clear();
    lcd.print("Enter Password:");
    inputPassword = "";
}

// Press A - reset OTP
else if (actual == 'A') {
    EEPROM.write(otpFlagAddress, 0);
    lcd.clear();
    lcd.print("OTP Reset Done");
    delay(2000);
    lcd.clear();
    lcd.print("Re-upload Code");
}

// inputs
else {
    inputPassword += actual;
    lcd.setCursor(0, 1);
    lcd.print(inputPassword);
}
}
}
```

7.3. Logic Verification

To verify logic accuracy, simulations were conducted in Arduino IDE and Proteus. The code performed reliably under different reset and input conditions.

8. Testing and Results

8.1. Performance Observations

- OTP delivery time: 3–5 seconds.
- Response time: ≤ 2 seconds.
- Power consumption: approx. 2A during GSM transmission.

8.2. Reliability

The system performed successfully during multiple test cycles, maintaining consistent GSM communication even with low signal strength.

8.3. Security Evaluation

The dynamic OTP generation makes brute-force and replay attacks infeasible. Each OTP becomes invalid after one use, ensuring system integrity.

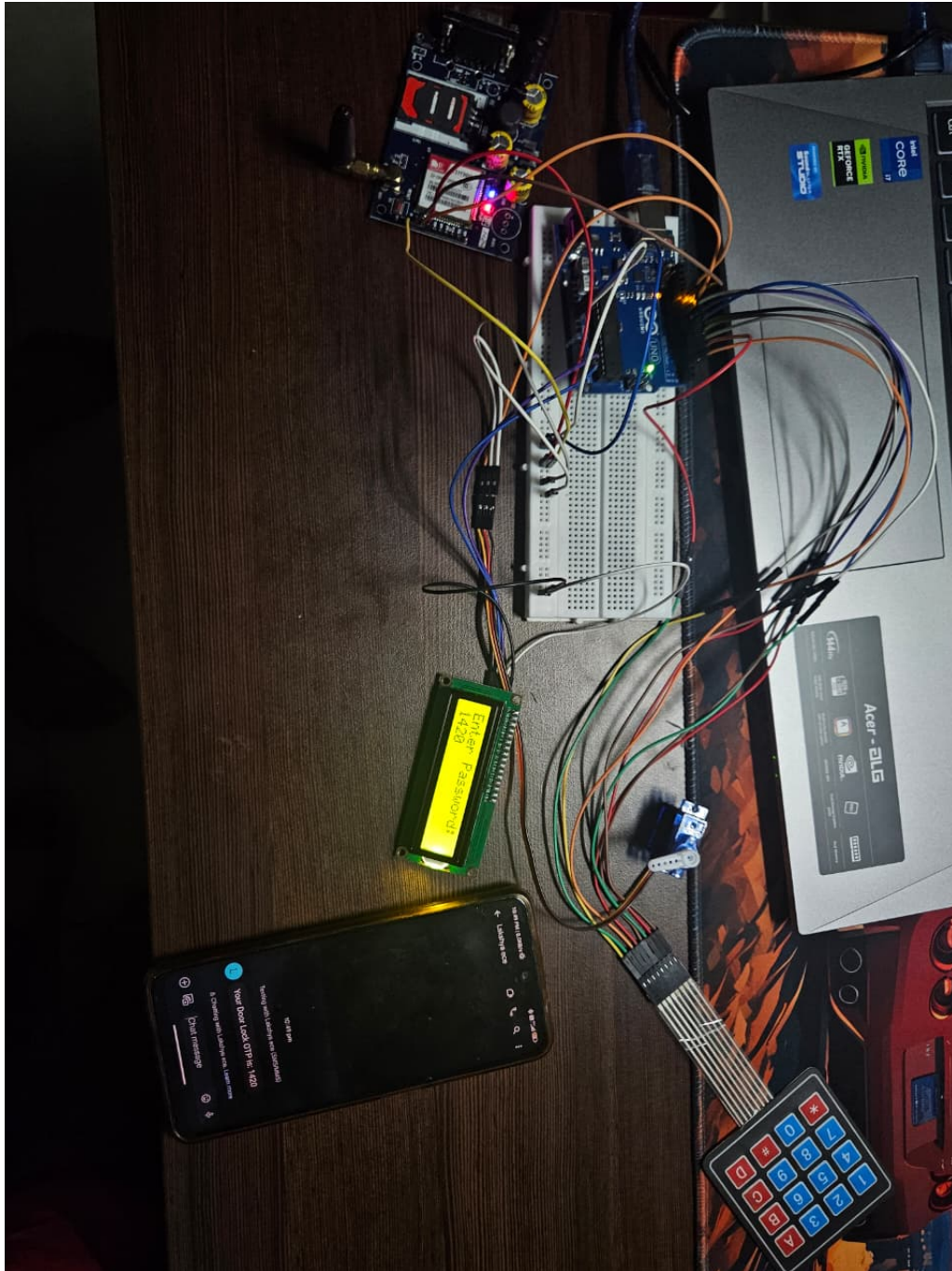


Figure 8.1: Photo of the Working Prototype Setup while entering OTP

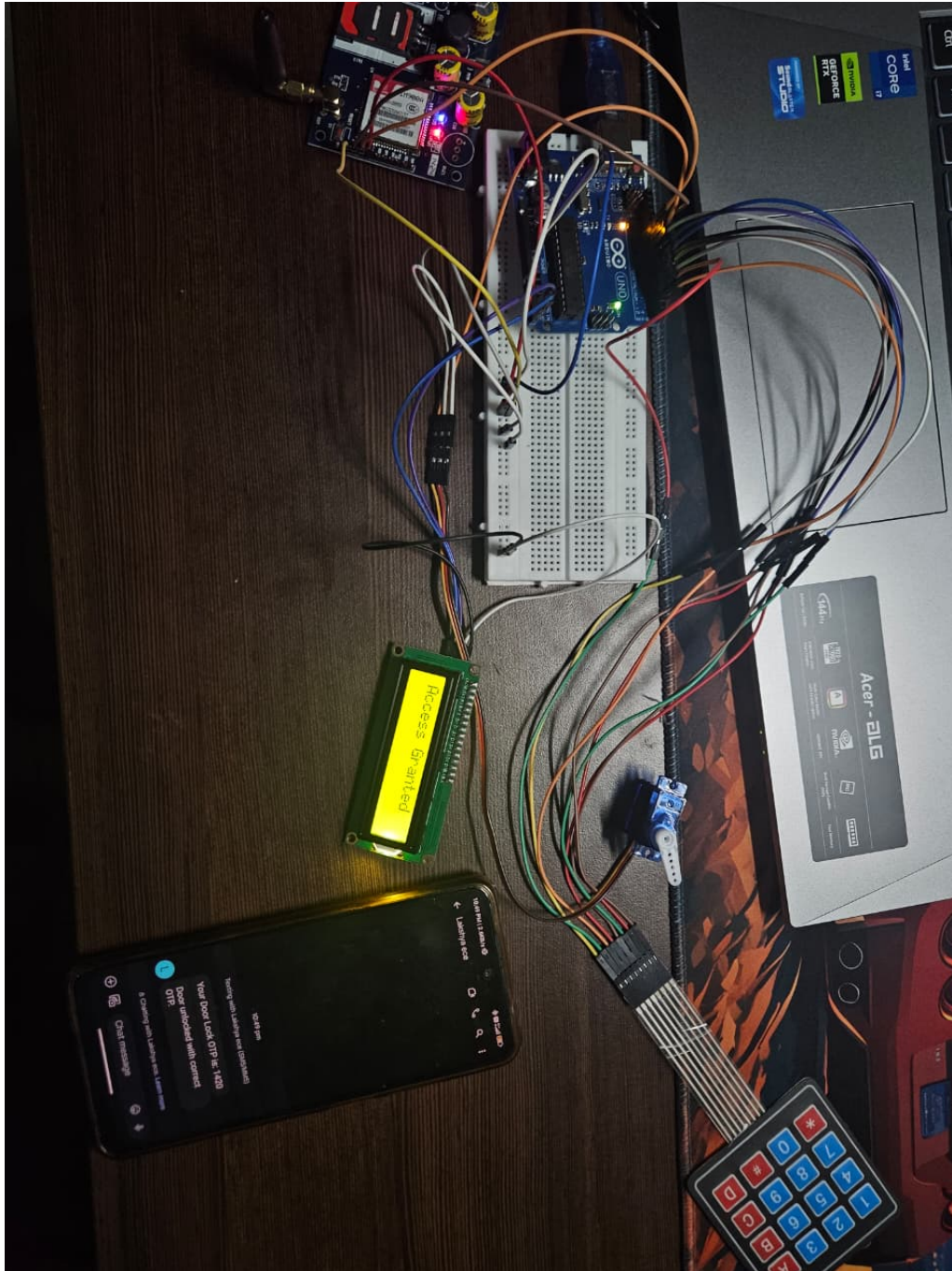


Figure 8.2: Photo of the Working Prototype Setup after entering OTP

9. Applications

- Home and apartment door security.
- Office access control.
- ATM and locker systems.
- Institutional or hostel entry systems.
- Warehouse and restricted area protection.

10. Future Scope

- Add biometric verification for multi-factor authentication.
- Include cloud-based logging and monitoring.
- Upgrade to LTE-based GSM for improved coverage.
- Integrate solenoid or magnetic locks for real deployment.
- Develop a mobile application interface.

11. Conclusion

This project demonstrates a secure and practical electric door lock system using GSM and OTP authentication. It effectively overcomes limitations found in traditional locks and network-dependent systems. By combining embedded control, real-time messaging, and dynamic password verification, the system ensures reliability and accessibility.

Its success in prototype testing suggests potential for commercialisation and integration into larger IoT-based security ecosystems.

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